WORKED SOLUTION

STEP III 1988, Question 9

Pure • Group Element Orders

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q9

The key idea

Part (i) is the division algorithm in disguise: write N as a multiple of n plus a remainder, and the minimality of n forces the remainder to vanish. Part (ii) works because conjugation by g respects products, so in a product of conjugates the inner $g^{-1}g$ pairs cancel. In part (iii) we apply conjugation by g again and again; since $g^5 = e$, five rounds take h back to itself, which pins down the order of h.

Solution

Part (i)

By the division algorithm we can write $N = qn + r$ with q an integer and $0 \leq r < n$. (This works for any integer N, including zero or negative ones.) Then, since $g^n = e$,

$$e = g^N = (g^n)^q g^r = e^q g^r = g^r.$$

If r were positive, it would be a positive integer smaller than n with $g^r = e$, which contradicts the definition of $n = o(g)$ as the *smallest* such integer. Hence $r = 0$, so $N = qn$ and n divides N.

Part (ii)

$m = 0$: both sides equal e.

$m \geq 1$: we use induction on m. The case $m = 1$ is immediate. If $gh^m g^{-1} = (ghg^{-1})^m$, then

$$\begin{aligned}(ghg^{-1})^{m+1} &= (ghg^{-1})^m (ghg^{-1}) = (gh^m g^{-1})(ghg^{-1}) \\ &= gh^m(g^{-1}g)hg^{-1} = gh^{m+1}g^{-1},\end{aligned}$$

which completes the induction. (Written out, $(ghg^{-1})(ghg^{-1})\cdots(ghg^{-1})$ telescopes: every $g^{-1}g$ in the middle cancels.)

$m \leq -1$: first, ghg^{-1} has inverse $gh^{-1}g^{-1}$, because

$$(ghg^{-1})(gh^{-1}g^{-1}) = gh(g^{-1}g)h^{-1}g^{-1} = ghh^{-1}g^{-1} = e.$$

Now write $m = -k$ with $k \geq 1$. Applying the positive case to the element h^{-1} in place of h,

$$gh^m g^{-1} = g(h^{-1})^k g^{-1} = (gh^{-1}g^{-1})^k = ((ghg^{-1})^{-1})^k = (ghg^{-1})^{-k} = (ghg^{-1})^m.$$

So the identity holds for every integer m.

Part (iii)

Using $ghg^{-1} = h^2$ and then part (ii) with $m = 2$,

$$g^2hg^{-2} = g\,(ghg^{-1})\,g^{-1} = gh^2g^{-1} = (ghg^{-1})^2 = (h^2)^2 = h^4,$$

as required.

The same step works at every stage. We claim that $g^khg^{-k} = h^{2^k}$ for all $k \geq 0$. This is true for $k = 0$, and if it holds for k, then by part (ii) with $m = 2^k$,

$$g^{k+1}hg^{-(k+1)} = g\,(g^khg^{-k})\,g^{-1} = gh^{2^k}g^{-1} = (ghg^{-1})^{2^k} = (h^2)^{2^k} = h^{2^{k+1}}.$$

Taking $k = 5$ and using $g^5 = e$ (so $g^{-5} = e$ too) gives

$$h = g^5hg^{-5} = h^{32}, \qquad \text{so} \qquad h^{31} = e$$

on multiplying by h^{-1}.

> **Ask yourself** We know $h^{31} = e$. Why does that make $o(h)$ exactly 31, and not just at most 31?

By part (i), $o(h)$ divides 31. Since 31 is prime, $o(h)$ is 1 or 31. But $o(h) = 1$ would mean $h = e$, which we are told is not the case. Therefore

$$o(h) = 31.$$

Such a group really exists, so the conditions are consistent. Take the permutations of $\{0, 1, \ldots, 30\}$ given by $h(x) = x + 1$ and $g(x) = 2x$, working modulo 31. Then $g^5(x) = 32x = x$, and $ghg^{-1}(x) = 2(\frac{x}{2} + 1) = x + 2 = h^2(x)$, and h does have order 31.

Checks

- The prime 31 makes sense: conjugating by g acts on the powers of h as "multiply the exponent by 2", and five doublings must return to the start, so $2^5 \equiv 1$ modulo $o(h)$. In other words, $o(h)$ divides $2^5 - 1 = 31$.
- A computer check with the permutations $g(x) = 2x$ and $h(x) = x + 1$ modulo 31 confirmed $g^5 = e$, $ghg^{-1} = h^2$, $g^khg^{-k} = h^{2^k}$ for $k = 0, \ldots, 5$, and $o(h) = 31$. It also tested (i) and (ii), for $-12 \leq m \leq 12$, on hundreds of random permutations.

Common slips

- In (ii), proving only the case $m \geq 1$. The question says *any integer*, so $m = 0$ and negative m need their own (short) argument.
- Concluding $o(h) = 31$ straight from $h^{31} = e$. That only gives $o(h) \mid 31$; you need the primality of 31 and the condition $h \neq e$ as well.

Beyond STEP

The map $x \mapsto gxg^{-1}$ is an *automorphism* of G (a structure-preserving bijection), and part (ii) is one consequence of that. Part (iii) is a small instance of a *semidirect product*: the group generated by g and h has order $5 \times 31 = 155$, and its existence depends on 2 having order 5 modulo 31. This construction is central to classifying groups of small order. Allenby's *Rings, Fields and Groups* is a good place to read about orders, conjugation and Lagrange's theorem.

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